

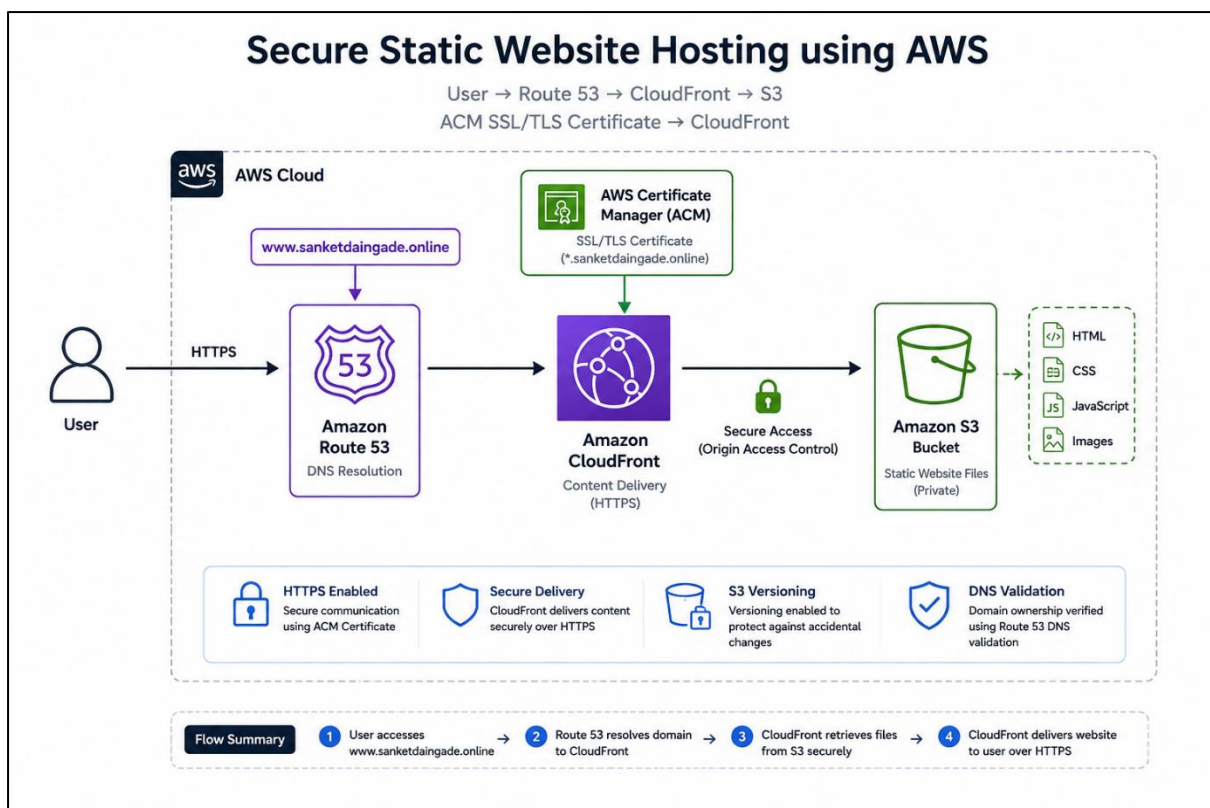
Secure Static Website Hosting using AWS S3, CloudFront & Route 53

Implementation Guide

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Project Overview

This project demonstrates secure static website hosting using **Amazon S3, CloudFront, ACM, and Route 53**. Website files are stored in an S3 bucket with versioning enabled. CloudFront delivers the website securely, while ACM provides the SSL/TLS certificate for HTTPS. Route 53 connects the custom domain `www.sanketdaingade.online` to CloudFront for secure website access.



Step 1 – Create and Configure S3 Bucket

Created an Amazon S3 bucket named **sanket-static-website-v1** to store the static website files. Bucket Versioning was enabled to maintain multiple versions of objects and protect against accidental changes or deletion. The required website files were then uploaded to the S3 bucket for hosting and delivery.

The screenshot shows the 'Create bucket' page in the AWS Management Console. The page is titled 'Create bucket' and includes a sub-header 'Buckets are containers for data stored in S3.' The 'General configuration' section is active, showing the 'AWS Region' as 'Asia Pacific (Mumbai) ap-south-1'. Under 'Bucket type', 'General purpose' is selected. The 'Bucket namespace' section shows 'Global namespace' selected. The 'Bucket name' field contains 'sanket-static-website-v1'. The 'Copy settings from existing bucket - optional' section is visible at the bottom.

Create bucket [info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region
Asia Pacific (Mumbai) ap-south-1

Bucket type [info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account. Regional namespaces are unique to your account. These buckets can never be created by another AWS account.

Bucket name [info](#)

sanket-static-website-v1

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

The screenshot shows the 'Upload' page in the AWS Management Console for the bucket 'sanket-static-website-v1'. The page is titled 'Files and folders (2 total, 2.2 KB)' and shows a table with two files: 'style.css' and 'index.html'. The 'Destination' section is expanded, showing the destination path 's3://sanket-static-website-v1'. The 'Permissions' and 'Properties' sections are also visible.

Files and folders (2 total, 2.2 KB)

All files and folders in this table will be uploaded.

☒	Name	Folder	Type	Size
☒	style.css	-	text/css	1.1 KB
☒	index.html	-	text/html	1.1 KB

Destination [info](#)

Destination
s3://sanket-static-website-v1 [info](#)

Destination details
Bucket settings that impact new objects stored in the specified destination.

Permissions
Grant public access and access to other AWS accounts.

Properties
Specify storage class, encryption settings, tags, and more.

[Cancel](#) [Upload](#)

Step 2 – Create SSL/TLS Certificate using AWS Certificate Manager

Created an SSL/TLS certificate using AWS Certificate Manager (ACM) in the **US East (N. Virginia)** region for ***.sanketdaingade.online** DNS validation was selected, and the required CNAME validation record was automatically created in Amazon Route 53. After successful domain validation, the certificate status changed from Pending validation to **Issued**, enabling secure HTTPS communication for the website.

The first screenshot shows the 'Request certificate' page in the AWS Certificate Manager console. The 'Certificate type' section has 'Request a public certificate' selected. The 'Domain names' section shows the domain '*.sanketdaingade.online' entered. The 'Allow export' section has 'Disable export' selected. The 'Validation method' section has 'DNS validation' selected.

The second screenshot shows the 'Create DNS records in Amazon Route 53' page. It displays a table with one record for the domain '*.sanketdaingade.online' with a validation status of 'Pending validation' and 'Is domain in Route 53?' set to 'Yes'.

The third screenshot shows the 'Certificate status' page for the certificate with identifier '25050fc6-e4de-4f0a-a504-918253705076'. The status is 'Issued'. The 'Domains' section shows a table with one record for the domain '*.sanketdaingade.online' with a status of 'Success'.

Domain	Status	Renewal status	Type	CNAME name	CNAME value
*.sanketdaingade.online	Success	-	CNAME	_b09ab25753072feeb5a5111f92e40f3b.sanketdaingade.online.	_9094e273157d04ccbd09187334d4a7ec.jkddztsvalidations.aws.

Step 3 – Create CloudFront Distribution

Created an Amazon CloudFront distribution named static-web-hosting to securely and efficiently deliver the static website content stored in Amazon S3. The custom domain www.sanketdaingade.online was added as an alternate domain name, and the ACM SSL/TLS certificate was attached to enable HTTPS access. The default root object was configured as index.html.

The screenshot displays the AWS CloudFront 'Create distribution' wizard, showing steps 1 through 4. The interface is in the 'Global' region, with the account name 'sanket_daingade (940831808963)' visible in the top right corner.

Step 1: Get started

Connect your websites, apps, files, video streams, and other content to CloudFront. We optimize the performance, reliability, and security for your web traffic.

Distribution options

Distribution name
Name will be stored as a tag on the resource. You can change the name, or more tags, later.

Description - optional

Distribution type

- ☒ **Single website or app**
Choose if each website or application will have a unique configuration.
- ☐ **Multi-tenant architecture - New**
Choose when you have multiple domains that need to share configurations. This is a common architecture for SaaS providers.

Step 2: Specify origin

Your origin is where your content (such as a website or app) lives. CloudFront works with AWS-based origins and origins hosted on other cloud providers.

Origin type

- ☒ **Amazon S3**
Deliver static assets like files and images, statically generated websites or single page applications (SPA).
- ☐ **Elastic Load Balancer**
Deliver applications hosted behind ELB such as dynamic websites, web services, and APIs.
- ☐ **API Gateway**
Deliver API endpoints for REST APIs hosted on API Gateway.
- ☐ **Elemental MediaPackage**
Deliver end-to-end live events or video on demand (VOD).
- ☐ **VPC origin**
Deliver applications and content hosted within private VPCs, such as EC2 instances and Application Load Balancers.
- ☐ **Other**
Refer to any AWS or non-AWS origin through its publicly resolvable URL.

Origin

S3 origin
Choose an AWS origin, or enter your origin's domain name. [Learn more](#)

[Browse S3](#)

Step 3: Enable security

Web Application Firewall (WAF)

- ☐ **Enable security protections**
Keep your application secure from the most common web threats and security vulnerabilities using AWS WAF. Blocked requests are stopped before they reach your web servers.
- ☒ **Do not enable security protections**
Select this option if your application does not need security protections from AWS WAF.

[Cancel](#) [Previous](#) [Next](#)

Step 4: Review and create

General configuration

Distribution name	Description	Billing
static-web-hosting	-	Pay-as-you-go (\$0/month)

[Edit](#)

Origin

[Edit](#)

S3 origin	Origin path	Grant CloudFront access to origin	Enable Origin Shield
sanket-static-website-v1.s3.ap-south-1.amazonaws.com	-	Yes	No

Connection attempts
3

Connection timeout
10

Cache settings

CloudFront will apply default cache settings tailored to serving content from a S3 origin. You can customize settings after you create your distribution.

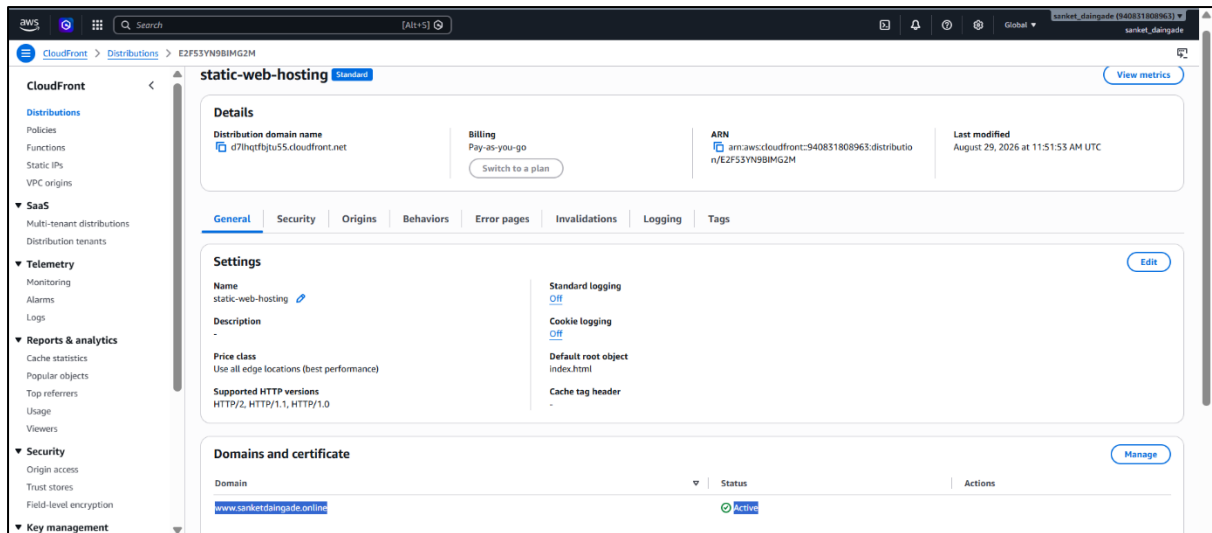
[Edit](#)

Security

Security protections	Use monitor mode	Use existing WAF configuration
None	No	No

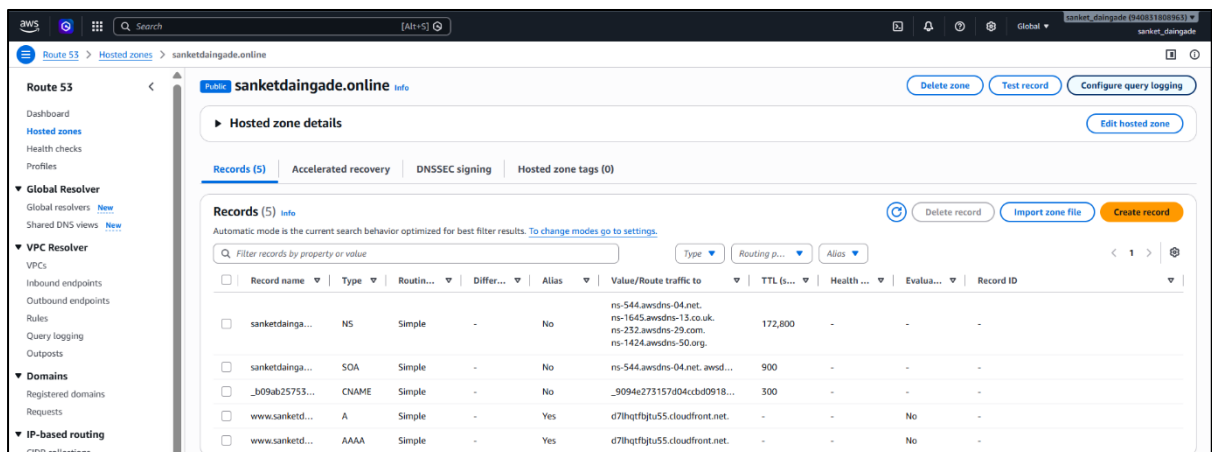
[Edit](#)

[Cancel](#) [Previous](#) [Create distribution](#)



Step 4 – Route Domain to CloudFront

Configured Amazon Route 53 to route `www.sanketdaingade.online` traffic to the CloudFront distribution. CloudFront routing was configured automatically, which created the required A and AAAA alias records in Route 53. This allows users to access the website through the custom domain over HTTPS.



Step 5 – Verify Website using Custom Domain

Finally, the deployment was verified by accessing <https://www.sanketdaingade.online> through a web browser. The website loaded successfully over a secure HTTPS connection, confirming that the S3 static website content, CloudFront distribution, SSL/TLS certificate, and Route 53 DNS configuration were working correctly.

